Notes on

Derived Categories

Typed by
Filippo Papallo

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Pisa

My original intention is to collect here some material about derived categories, with applications to algebraic geometry, which will hopefully turn out to be useful for my thesis. I will mainly follow the first chapters of the book "*Fourier-Mukai transforms in algebraic geometry*" by Daniel Huybrects [Huy06].

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List of Lectures

Chapter I.

Structure of the thesis

I

The idea is to go through the article *Braid groups actions on derived categories of coherent sheaves*, by Siedel and Thomas.

The article starts with a sections containing motivations on the theory: the authors give a survey on the main fields where the derived category $\mathbf{D}^b(X)$ appears. In particular, symplectic geometry and mirror symmetry are the main interest of the paper.

Section 2 introduces braid group actions on a category. For a detailed description, check **Deligne**'s article "*Action du group de tresses sur une catégorie*". In Section 2.b spherical objects and twist functors are introduced in a pure functorial way, i.e. using only monoidal abelian categories. By introducing enough hypothesis on our category, we can study braid relations in a pure abstract way: Section 2.c is devoted to the contruction of braid relations between twist functors induced by spherical objects. Here, **Proposition 2.12** and **Proposition 2.13** show that there exists a braid group action an suitable category $\mathcal{K}$. The main result is stated in **Theorem 2.18**.

Theorem. — *Suppose that $n \geq 2$. Then the homomorphism*

$$\rho : B_{m+1} \longrightarrow Autoeq(\mathbf{D}^b(X))$$

is injective, and in fact the following stronger statement holds: if $g \in B_{m+1}$ is not the identity element, then $\rho(g)(E_i) \neq E_i$ for some $i \in \{1, ..., m\}$.

The proof of the faithfulness of this action is the whole Section 4. The machinery involved is mainly based on the theory of **differential graded algebras** (dgas). A detailed survey of dgas can be found in **Bernstein** and **Lunts**'s "*Equivariant sheaves and functors*".

In the context of the derived category of a dga, standard twist functors are introduced. These twists commute with equivalences induced by qis of dgas. **Lemma 4.3** relates the standard twist functors t of dgas with the twist functors T_E in our favorite triangulated category.

We then try to study the twist functors T_{E_i} by analizing the dga $end(E)$. We will focus on a particular class of dgas, namely **intrinsically formal** graded algebras: these objects are determined by their cohomology. This property can be studied in terms of the vanishing of some **Hochschild cohomology** modules, as stated in **Theorem 4.7**. A simple intoduction to this topic can be found in the "Istituzioni di Algebra" notes by **Tamas**.

The proof of this fact is by no mean trivial. The authors develop a theory of A_∞**-morphisms**, because they induce morphisms of dgas.

In Section 4.c the algebras $A_{m,n}$ are introduced as the path algebras of a suitable **quiver**. I should check for a little survey on this topic. This section aims to explain how to translate

the problem on twist funtors with the language of dgas, indeed $A_{m,n}$ are isomorphic to some $end(E)$.

Standard twist functors $t_i : \mathbf{D}(A_{m,n}) \to \mathbf{D}(A_{m,n})$ are exact equivalences (**Lemma 4.11**) that satisfy the braid relations.

Finally, the faithfullness of the braid action is proved in Section 4.d for the derived categories of the algebras $A_{m,n}$. To achieve this, one studies the twist functor associated to the generator of the center of B_{m+1}.

The authors conjecture that $(t_1 t_2 \dots t_m)^{m+1}$ is isomorphic to the translation $[2m - (m+1)n]$.

The proof of the faithfulness relies on topology: in particular, **mapping class groups** of the punctured disc are involved. It is well known their link with the braid groups, and **Lemma 4.16** explains the link between some diffeomorphisms and the generator of the center of B_{m+1}.

Geometric intersection numbers are defined; one can check "*Quivers, Floer cohomology, and braid group actions*" by **Khovanov** and **Seidel** for an insight. In fact, we can find braid group actions already in this paper; Siedel and Thomas make a comparison with their definition $A_{m,n}$.

Strictly speaking, Section 4.d is a comparison with the article by Khovanov and Seidel, by the end of which the faithfulness is proved in the category of some dga.

The final section is dedicated to the computation of Hochschild cohomology of $A_{m,n}$, so that its intrinsic formality is proved. This fact is essential in the proof of **Theorem 2.18**, together with the faithfulness described above.

Study plan

I'm new to the following topics:

- derived category of a scheme;
- Fourier-Moukai Transforms;
- group actions on a category;
- DG-algebras;
- Hochschild cohomology and its applications;
- quivers and representations;
- mapping class group.

Thus, I'll need some study!

Structure idea

Here I write a possible structure of the strategy:

- introduction, which will be written at the very end, possibly with motivations;

Structure idea

- recollection of some properties of derived categories;
- a survey about DG-algebras;
- something about quivers and their path algebras;
- we start to set the main tools: definition of spherical objects in a triangulated category, and definition of twists;
- definition of group action over a category. Braids acting on a category and faithfulness: discussion and proof, assuming **Theorems 4.13** e **4.21**, whose proofs are postponed;
- technicalities about dgas, intrinsic formality and geometric intersections;
- applications to the algebro-geometric context (definition of FMT is needed) and examples; recent develpments.

Chapter II.

Differential graded algebras

II

For the whole chapter, fix a commutative ring Λ with unit $1 \neq 0$. We are going to study new additional structures on Λ-algebras, namely *differential* and *graded* structures; the aim of this is to define and understand their main properties, and after that we will be able to approach the related concept of graded modules over these algebras. In the geometric setting, our main interest will be the special case of the semisimple algebra $\Lambda = k$, for some field and $m \leq 1$.

The differential structure on these new objects allows us to talk about *cohomology*. It is well known that isomorphic complexes yield the same information in cohomology, while the converse is false in general; thus, we are going to focus on a class of very well-behaved graded algebras, *intrinsically formal* algebras: strictly speaking, these objects are determined by their cohomology.

Intrinsic formality of DGA-algebras can be characterized in terms of vanising of some *Hochschild cohomology groups*, thus we safe a section to define this notion.

DG-algebras

Every Λ-algebra in this chapter is associative, not necessarily commutative, with unit $1 \neq 0$.

Definition. — A Λ-algebra A is **graded algebra** if there exist submodules $\{ A^k \mid k \in \mathbb{Z} \}$, such that $1 \in A^0$ and, for every $i, j \in \mathbb{Z}$, we have multiplications

$$A^i \otimes_\Lambda A^j \longrightarrow A^{i+j}\,, \quad x \otimes y \longmapsto xy\,,$$

such that $A = \bigoplus_{k \in \mathbb{Z}} A^k$. An element of A^k is said **homogeneous of degree** k; sometimes we will simply write $\deg(x)$ instead of k, for a homogeneous element $x \in A^k$.

A graded Λ-algebra is **graded-commutative** (or **anti-**, sometimes called **super-**commutative), if it holds

$$yx = (-1)^{kh} xy\,, \quad \text{for every } x \in A^k, y \in A^h\,.$$

The concept of differential graded algebra A is not new: it boils down to a cochain complex whose differential must satisfy some compatibility axiom with respect to the multiplication.

Definition. — A **differential** on a graded algebra A is a Λ-linear endomorphism $d : A \to A$ such that $d^2 = 0$ and, for every $k \in \mathbb{Z}$, it holds $d(A^k) \subset A^{k+1}$; moreover, d must satisfy the following **graded Liebniz rule**:

$$d(xy) = (dx)y + (-1)^k x(dy)\,, \quad \text{for } x \in A^k\,. \tag{II.0.1}$$

A **DG-algebra** $\mathcal{A} = (A, d)$ is a graded Λ-algebra A endowed with a differential d.

Remark. — Notice that (II.0.1) implies that $d(1) = 0$.

The condition $d(A^k) \subset A^{k+1}$ in the above definition allows us to interpret a DG-algebra $\mathcal{A} = (A, d)$ as a sequence

$$\begin{tikzcd} \dots \arrow[r] & A^{k-1} \arrow[r, "d"] \arrow[rr, bend right, "0"'] & A^k \arrow[r, "d"] & A^{k+1} \arrow[r] & \dots \end{tikzcd}$$

thus, any DG-algebra is a cochain complex $\mathcal{A} = A^\bullet$ of Λ-modules, whose coboundary map is the same d at each level; this means we can compute its cohomology $H^*(\mathcal{A})$. The interesting fact is that this is not just a module: indeed, the relation (II.0.1) implies that the multiplication induced on $H^*(\mathcal{A})$ is well defined and moreover

$$H^k(\mathcal{A}) \otimes H^h(\mathcal{A}) \longrightarrow H^{k+h}(\mathcal{A}) \,, \quad [x] \cdot [y] = [xy]$$

shows that the cohomology inherits a graded Λ-algebra structure $H^*(\mathcal{A}) = \oplus_k H^k(\mathcal{A})$. By endowing it with a trivial differential $d = 0$, we conclude that the cohomology $H^*(\mathcal{A})$ of a DG-algebra $\mathcal{A}$ is again a DG-algebra.

We define $\mathbf{DG}_\Lambda$ to be the category whose objects are DG-algebras over Λ, and morphisms $f : \mathcal{A} \to \mathcal{B}$ between them given by homomorphisms of unital Λ-algebras

$$f : A \longrightarrow B \,, \quad \text{s.t. } f(1_A) = 1_B \,,$$

which are also maps of complexes, that is $f(A^k) \subset B^k$ and $f \circ d = \delta \circ f$. One can check that $\mathbf{DG}_\Lambda$ is an abelian category (we might expect this because it is a category of modules).

Definition. — Given a DG-algebra $\mathcal{A} = (A, d)$, we define its **opposite algebra** $\mathcal{A}^{op} = (A^{op}, d)$ as the DG-algebra whose elements and differential are the same of $\mathcal{A}$, but we consider a new multiplication

$$a \cdot^{op} b := (-1)^{\deg a \deg b} ba \,,$$

where ba denotes the usual multiplication in $\mathcal{A}$. Notice that $\mathcal{A}$ is graded-commutative if and only if $\mathcal{A} = \mathcal{A}^{op}$.

Example. — Let X be a smooth real manifold. The algebra $\Omega^\bullet(X)$ of smooth differential forms endowed with the *exterior derivative* is a DG-algebra and $\Omega^\bullet$ determines a functor

$$\Omega^\bullet : \mathbf{Man} \longrightarrow \mathbf{DG}_\mathbb{R} \,,$$

where **Man** is the category of smooth manifolds and smooth maps.

Example. — Taking cohomology defines a covariant functor $H^* : \mathbf{DG}_\Lambda \to \mathbf{DG}_\Lambda$.

Example. — Given a finite dimensional k-vector space V, define $T^0 := k$ and for each $k \geq 1$ set

$$T^k(V) := T^{\otimes k} = \underbrace{V \otimes_k \cdots \otimes_k V}_{k \text{ times}} \,.$$

Then $T(V) := \bigoplus_{k\geq 0} T^k(V)$ naturally inherits an associative multiplication

$$T^k(V) \otimes_k T^h(V) \longrightarrow T^{k+h}(V)\,;$$

from the canonical isomorphism $k \otimes_k V \simeq V$ we deduce that $1 \in T^0(V)$ is the unit of the multiplication, hence $T(V)$ is a graded algebra over k, called **tensor algebra** of V.

Let $\{e_1, \dots, e_n\}$ be a basis of V, and set $A^{-k} := T^k(V)$. We can define a differential $d : T(V) \to T(V)$ by setting on basis elements

$$d(e_i) = (-1)^i\,, \quad i \in \{\,1, 2, \dots, n\,\}\,,$$

and then extending it componentwise to a map

$$d : A^{-k} \longrightarrow A^{-k+1}\,, \quad d(e_{i_1} \otimes \cdots \otimes e_{i_k}) = \sum_{j=1}^{k} e_{i_1} \otimes \cdots \otimes d(e_{i_j}) \otimes \cdots \otimes e_{i_k}\,.$$

From now on, we will consider DG-algebras with terms of non-negative degree, that is $A^k = \mathbf{0}$ for $k < 0$, and set $\mathcal{A}^+ := \bigoplus_{k\geq 0} A^k$ for the submodule of positive degree elements.

Definition. — An **augmentation** on a DG-algebra $\mathcal{A}$ is a morphism $\varepsilon : \mathcal{A} \to \Lambda$ of unital Λ-algebras such that $\varepsilon \circ d = 0$ and

$$\varepsilon(x) = 0\,, \quad \text{for } x \in \mathcal{A}^+\,.$$

Thus, an augmentation is determined by its restriction to A^0. If we write $\iota : \Lambda \to A$ be the map $\iota(\lambda) := \lambda \cdot 1_A$, then $\varepsilon \circ \iota = \mathbf{1}_\Lambda$ and hence we may identify the ground ring Λ with a subring of A; in particular, notice that Λ is a direct summand of A by the Split Lemma. Notice that the condition $\varepsilon d = 0$ implies that there is a well defined aumentation on $H^*(\mathcal{A})$.

Definition. — A DG-algebra $\mathcal{A}$ endowed with an augmentation ε will be called **DGA-algebra** (which stands for **differential augmented graded algebra**). Given two DGA-algebras $\mathcal{A} = (A, d, \varepsilon)$ and $\mathcal{A}' = (A', d', \varepsilon')$, a **DGA-homomorphism** is a morphism of DG-algebras compatible with augmentations, that is $\varepsilon' \circ f = \varepsilon$.

$$\begin{tikzcd} A \arrow[rr, "f"] \arrow[dr, "\varepsilon"'] & & A' \arrow[dl, "\varepsilon'"] \\ & \Lambda & \end{tikzcd}\,.$$

We will write $\mathbf{DGA}_\Lambda$ for the category of DGA-algebras over Λ. Notice that a DGA-homomorphism induces a morphism $f* : H*(\mathcal{A}) \to H^*(\mathcal{A}')$ of DGA-algebras, hence cohomology defines a covariant functor $H^* : \mathbf{DGA}_\Lambda \to \mathbf{DGA}_\Lambda$.

Example. — Given a topological space X, let $S(X)$ denote its singular simplicial complex. For each pair of spaces X, Y, one can define a chain map $\sigma : S(X) \otimes_{\mathbb{Z}} S(Y) \to S(X \times Y)$. Now, assume X is a topological group, with a continuous multiplication

$$\mu : X \times X \longrightarrow X\,,$$

with neutral element $e \in X$; then the composition

$$S(X) \otimes_{\mathbb{Z}} S(X) \xrightarrow{\sigma} S(X \times X) \xrightarrow{\mu_\bullet} S(X)$$

defines an associative multiplication on $S(X)$, whose neutral element is the 0-simplex e, called **Pontrjagin product**. Moreover, there exists an augmentation $\varepsilon : S(X) \to \mathbb{Z}$ which sends each 0-simplex (i.e. a point of X) to $1 \in \mathbb{Z}$, and vanishes on higher dimensional simplices; this makes $S(X)$ into a DGA-algebra[1].

II.1. Proposition. — *The category* $\mathbf{DGA}_\Lambda$ *is a monoidal category.*

Proof. Consider any two DGA-algebras $\mathcal{A} = (A^\bullet, d, \varepsilon)$ and $\mathcal{B} = (B^\bullet, \delta, \eta)$. We define their tensor product $\mathcal{A} \otimes \mathcal{B}$ to be the cochain complex $A^\bullet \otimes B^\bullet$ as defined in Example **??**, i.e. the complex whose degree n elements are

$$(A^\bullet \otimes B^\bullet)^n = \bigoplus_{k \in \mathbb{Z}} A^k \otimes_\Lambda B^{n-k}\,,$$

equipped with the differential D defined by

$$D(a \otimes b) = da \otimes b + (-1)^{\deg(a)} a \otimes \delta b\,.$$

We can define the multiplication

$$(a \otimes b) \cdot (a' \otimes b') := (-1)^{\deg(a)\deg(a')}(aa') \otimes (bb')\,,$$

in such a way that $(A^\bullet \otimes B^\bullet, D)$ becomes a DG-algebra: indeed, it holds

$$\begin{aligned}
D\big((a \otimes b) \cdot (a' \otimes b')\big) &= (-1)^{\deg(a)\deg(a')} D\big((aa') \otimes (bb')\big) \\
&= (-1)^{\deg(a)\deg(a')} \big(d(aa') \otimes bb' + (-1)^{\deg(aa')} aa' \otimes \delta(bb')\big) \\
&= (-1)^{\deg(a)\deg(a')} \Big((da)a' \otimes bb' + (-1)^{\deg(a)} a(da') \otimes bb' \\
&+ (-1)^{\deg(a)+\deg(a')} \big(aa' \otimes (\delta b)b' + (-1)^{\deg(b)} aa' \otimes b(\delta b')\big)\Big) \\
&= \big((-1)^{\deg(a')}(da \otimes b + (-1)^{\deg(a)} a \otimes \delta b) \cdot (a' \otimes b')\big) \\
&+ \big((a \otimes b) \cdot (da' \otimes b' + a' \otimes \delta b')\big) \\
&= (-1)^{\deg(a')} \big(D(a \otimes b) \cdot (a' \otimes b') + \big) \\
&= D(a \otimes b) \cdot (a' \otimes b') + (-1)^{\deg(a \otimes b)} (a \otimes b) \cdot D(a' \otimes b')
\end{aligned}$$

Please fix this.

so the differential satisfies the Liebniz rule (II.0.1). Finally, the map α defined on every simple tensor $a \otimes b$ by

$$\alpha(a \otimes b) := \varepsilon(a)\eta(b) = 0\,,$$

is an augmentation on $\mathcal{A} \otimes \mathcal{B}$ because it clearly vanishes on elements of $(\mathcal{A} \otimes \mathcal{B})^0 = A^0 \otimes B^0$, and by composing it with the differential one gets

$$\alpha \circ D = (\varepsilon \circ d)\eta \pm \varepsilon(\eta \circ \delta) = 0\,.$$

This shows that $\mathcal{A} \otimes \mathcal{B} = (A^\bullet \otimes B^\bullet, D, \alpha)$ is a DGA-algebra. One can check that $\mathbf{DG}_\Lambda$ equipped with $\otimes$ becomes a monoidal category by an analgous argument as for $C^\bullet(\mathbf{Mod}_\Lambda)$. □

[1] In this case, $S(X)$ is a DGA-algebra whose differential follows the *homological* convention, while our definition is based on the cochain complex convention.

DG-modules

Definition. — Let $\mathcal{A} = (A, d)$ be a DG-algebra. A **left DG-module** $\mathcal{M}$ over $\mathcal{A}$, or simply **$\mathcal{A}$-DG-modules**, is a cochain complex $\mathcal{M} = (M^\bullet, d_M)$ in $C^\bullet(\mathbf{Mod}_A)$ such that:

- the multiplication by an homogeneous element $a \in A^k$ is a group homomorphism of degree k, that is
$$A^k \otimes_{\mathcal{A}} M^h \longrightarrow M^{k+h}\,, a \otimes m \longmapsto am\,;$$
- the differential d_M satisfies the following Liebniz rule:
$$d_M(am) = (da)m + (-1)^{\deg(a)} a(d_M m)\,, \quad \text{for } a \in \mathcal{A}, m \in \mathcal{M}\,. \tag{II.1.1}$$

A **morphism of $\mathcal{A}$-DG-modules** $f : \mathcal{M} \to \mathcal{N}$ is simply a cochain map $f : M^\bullet \to N^\bullet$ of complexes of A-modules.

One defines a **right DG-module** $\mathcal{M}$ over $\mathcal{A}$ (in short a **DG-$\mathcal{A}$-module**) in an analogous way, but the Liebniz rule for the right multiplication becomes

$$d_M(ma) = (d_M m)a + (-1)^{\deg m} m(da)\,, \quad \text{for } a \in \mathcal{A}, m \in \mathcal{M}\,.$$

Example. — Left DG-modules over a fixed DG-algebra $\mathcal{A}$, together with maps of complexes, form a category called ${}_{\mathcal{A}}\mathbf{DGMod}$. If $\mathcal{A} = A^0$, then there is no differential structure on the ground ring, thus an $\mathcal{A}$-DG-modules is just a cochain complex of A^0-modules. In particular, ${}_{\mathbb{Z}}\mathbf{DGMod} = C^\bullet(\mathbf{Ab})$.

Example. — Given any $\mathcal{A}$-DG-module $\mathcal{M}$, its cohomology $H^*(\mathcal{M})$ is naturally a cochain complex (with trivial differential), which has a natural module structure over $H^*(\mathcal{A})$.

From now on we borrow all the terminology used for complexes of modules. For example, we say that a $\mathcal{A}$-DG-module $\mathcal{M}$ is **acyclic** if $H^*(\mathcal{M}) = \mathbf{0}$; we say that a morphism $f : \mathcal{M} \to \mathcal{N}$ of $\mathcal{A}$-DG-modules is a **quasi-isomorphism** if it induces isomorphisms $f^* : H^*(\mathcal{M}) \simeq H^*(\mathcal{N})$, and so on and so forth.

Definition. — We define the **translation functor** $[1] : {}_{\mathcal{A}}\mathbf{DGMod} \to {}_{\mathcal{A}}\mathbf{DGMod}$ by sending any $\mathcal{A}$-DG-module $\mathcal{M}$ to the graded module

$$(\mathcal{M}[1])^n := M^{n+1}\,, \quad d_{M[1]} := -d_M\,,$$

in which the A-module structure on $M[1]$ is **twisted**, i.e. we define the scalar multiplication as
$$a * m := (-1)^{\deg(a)} am\,,$$
where am is the multiplication in M.

At this point we have enough structure to talk about *triangles*: in fact, our next goal is to develop enough theory to be able to state and prove that the homotopy category $\mathbf{K}({}_{\mathcal{A}}\mathbf{DGMod})$ is a triangulated category.

Definition. — Two morphisms $f, g : \mathcal{M} \to \mathcal{N}$ in ${}_{\mathcal{A}}\mathbf{DGMod}$ are **homotopic** if there exists a homotopy $s : \mathcal{M} \to \mathcal{N}[-1]$ of A-modules (possibly **not** of $\mathcal{A}$-DG-modules) such that

$$f - g = sd_m + d_N s\,,$$

in which case we write $f \sim g$.

The article by Seidel and Thomas uses the derived category of a DG-algebra, stating that it is **not** the same thing as the derived catgory of an abelian category (see **Warning** at

Since null-homotopic morphisms form a 2-sided ideal in $\mathrm{Hom}_{{}_{\mathcal{A}}\mathbf{DGMod}}(\mathcal{M}, \mathcal{N})$, we may quotient by the equivalence relation $f \sim g$ given by homotopy to obtain the **homotopy category** $\mathbf{K}({}_{\mathcal{A}}\mathbf{DGMod})$, whose objects are $\mathcal{A}$-DG-modules and morphisms between them are

$$\mathrm{Hom}_{\mathbf{K}({}_{\mathcal{A}}\mathbf{DGMod})}(\mathcal{M}, \mathcal{N}) := \mathrm{Hom}_{{}_{\mathcal{A}}\mathbf{DGMod}}(\mathcal{M}, \mathcal{N}) / \sim\ .$$

Given a morphism $f : \mathcal{M} \to \mathcal{N}$ in ${}_{\mathcal{A}}\mathbf{DGMod}$, we may build the **cone** of f in the usual way, namely $\mathbf{C}(f) := N \oplus M[1]$ with differential $(d_N + f, -d_M)$; this naturally inherits a structure of $\mathcal{A}$-DG-module, thus we may define a **strict triangle**

Is it distinguished? Stack Project uses a different class of distinguished triangles...

$$\mathcal{M} \xrightarrow{f} \mathcal{N} \longrightarrow \mathbf{C}(f) \longrightarrow \mathcal{M}[1]\,.$$

An **exact triangle** in $\mathbf{K}({}_{\mathcal{A}}\mathbf{DGMod})$ is a diagram $\mathcal{M}' \to \mathcal{N}' \to \mathcal{K}' \to \mathcal{M}'[1]$ which is isomorphic (in he homotopy category) to a strict triangle as above: explicitly, there is a diagram

$$\begin{tikzcd}
\mathcal{M}' \arrow[r] \arrow[d, "\sim"] & \mathcal{N}' \arrow[r] \arrow[d, "\sim"] & \mathcal{K}' \arrow[r] \arrow[d, "\sim"] & \mathcal{M}'[1] \arrow[d, "\sim"] \\
\mathcal{M} \arrow[r, "f"] & \mathcal{N} \arrow[r] & \mathbf{C}(f) \arrow[r] & \mathcal{M}[1]\,,
\end{tikzcd}$$

which commutes up to homotopy, in which vertical maps are homotopy equivalences.

As it happens for the homotopy category $\mathbf{K}(\mathcal{A})$ of an abelian category $\mathcal{A}$, it turns out that $\mathbf{K}({}_{\mathcal{A}}\mathbf{DGMod})$ becomes a triangulated category if we equip it with the translation functor $[1]$ and take the exact triangles as the family of distinguished triangles. In this category, quasi-isomorphisms of $\mathcal{A}$-DG-modules form a *multiplicative system*, and hence we can build the **derived category** $\mathbf{D}({}_{\mathcal{A}}\mathbf{DGMod})$ by describing morphisms as *roofs*; this category also inherits a triangulated structure.

II.2. Theorem. — *If $\Phi : \mathcal{A} \to \mathcal{B}$ is a quasi-isomorphism of DG-algebras, then the* "restriction of scalars" *induces an exact equivalence $\Phi^* : \mathbf{D}({}_{\mathcal{B}}\mathbf{DGMod}) \to \mathbf{D}({}_{\mathcal{A}}\mathbf{DGMod})$ of triangulated categories.*

It is explained in Equivariant Sheaves.

Hochschild (co)homology

For now, let Λ be a commutative unital ring and consider A an associative Λ-algebra with unit 1_A, which is not necessarily commutative. As always, we identify the two multiplicative identities via the structure map $\lambda \in \Lambda \mapsto \lambda \cdot 1_A$.

Definition. — We define the **opposite ring** A^{op} to be the Λ-module A, endowed with the multiplication $a \cdot^{op} b := ba$. The **enveloping algebra** A^e is the Λ-algebra $A \otimes_\Lambda A^{op}$, where the multiplication is defined on simple tensors by

$$(a_1 \otimes b_1) \cdot^e (a_2 \otimes b_2) := a_1 a_2 \otimes b_2 b_1 \,.$$

Remark. — Notice that $A = A^{op}$ whenever A is commutative.

Definition. — Given two commutative unital rings B and R, a **B-R-bimodule** is an abelian group which is both a left B-module and a right R-module in such a way that the two structures are compatible, i.e. it holds

$$(bm)r = b(mr)\,, \quad \text{for every } b \in B, m \in M, r \in R\,.$$

Notice that an A-A-bimodule structure on M is the same as a left A^e-module, in which the scalar multiplication is given by $(a_1 \otimes a_2) \cdot m = a_1 m a_2$. For instance, if we consider M to be the n-fold tensor product

$$A^{\otimes n} := \underbrace{A \otimes_\Lambda \cdots \otimes_\Lambda A}_{n \text{ times}}\,,$$

then its A^e-structure is given by

$$(b \otimes c) \cdot (a_1 \otimes \cdots \otimes a_n) = b a_1 \otimes \cdots \otimes a_n c\,.$$

Definition. — The **bar resolution** of A is the sequence of A-A-bimodules

$$\mathrm{Bar}(A) : \ldots \longrightarrow A^{\otimes 4} \xrightarrow{d_2} A^{\otimes 3} \xrightarrow{d_1} A^{\otimes 2}\,,$$

where $d_n : A^{\otimes n+2} \to A^{\otimes n+1}$ is defined for every $n \geq 0$ by

$$d_n(a_0 \otimes \cdots \otimes a_{n+1}) = \sum_{i=0}^{n} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1}\,. \tag{II.2.1}$$

II.3. Lemma. — *Let $d_0 : A \otimes A \to A$ be the multiplication $d_0(a \otimes b) = ab$. Then the $\mathrm{Bar}(A) \to A \to 0$ is an acyclic complex.*

Proof. For every $n \geq -1$, we define the Λ-linear map

$$s_n : A^{\otimes n+2} \longmapsto A^{\otimes n+1}\,, \quad s_n\,(a_0 \otimes \cdots \otimes a_{n+1}) = 1 \otimes \otimes \cdots \otimes a_{n+1}\,.$$

To conclude, it is enough to check that $s_\bullet$ defines a contracting homotopy, i.e. $ds + sd$ is the identity at each level. For every simple tensor one computes

$$\begin{aligned}
d_{n+1} s_n\,(a_0 \otimes \cdots \otimes a_{n+1}) &= d_{n+1}\,(1 \otimes a_0 \otimes \cdots \otimes a_{n+1}) \\
&= a_0 \otimes \cdots \otimes a_{n+1} + \sum_{i=0}^{n} (-1)^{i+1} \otimes a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1} \\
&= a_0 \otimes \cdots \otimes a_{n+1} - s_{n-1} \left(\sum_{i=0}^{n} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1} \right) \\
&= a_0 \otimes \cdots \otimes a_{n+1} - s_{n-1} d_n (a_0 \otimes \cdots \otimes a_{n+1})\,.
\end{aligned}$$

□

Thus, the bar resolution is a chain complex in $C_\bullet({}_{A^e}\mathbf{Mod})$ which is quasi-isomorphic to $A[0]$.

Definition. — Let M be an A-A-bimodule. The **Hochschild chain complex with coefficients in** M is the complex

$$C_\bullet(A;M) := M \otimes_{A^e} \mathrm{Bar}(A)\,.$$

Dually, we define the **Hochschild cochains with coefficients in** M to be the complex

$$C^\bullet(A;M) := \mathrm{Hom}_{A^e}(\mathrm{Bar}(A), M)\,.$$

More explicitly, for every $n \geq 0$ we see there is an isomorphism of left A^e-modules

$$\begin{aligned} C_n(A;M) = M \otimes_{A^e} A^{\otimes(n+2)} &\xrightarrow{\sim} M \otimes_\Lambda A^{\otimes n}\,, \\ m \otimes_{A^e} (a_0 \otimes a_1 \otimes \cdots \otimes a_n \otimes a_{n+1}) &\longmapsto a_{n+1} m a_0 \otimes a_1 \otimes \cdots \otimes a_n\,, \end{aligned}$$

were we declare $A^{\otimes 0} = \Lambda$; thus the induced boundary δ map looks like

$$\begin{aligned} \delta_n\left(m \otimes a_1 \otimes \cdots \otimes a_n\right) = {} & m a_1 \otimes \cdots \otimes a_n \\ & + m \otimes d_{n-2}\left(a_1 \otimes \cdots \otimes a_n\right) \\ & + (-1)^n a_n m \otimes a_1 \otimes \cdots \otimes a_{n-1}\,. \end{aligned}$$

Similarly, one can find that the n-cochains are

$$\mathrm{Hom}_{A^e}\left(A^{\otimes(n+2)}, M\right) \simeq \mathrm{Hom}_\Lambda\left(A^{\otimes n}, M\right)\,, \tag{II.3.1}$$

thus we get the coboundary maps

$$\begin{aligned} \partial_n f\left(a_1 \otimes \cdots \otimes a_n\right) = {} & a_1 f\left(a_2 \otimes \cdots \otimes a_n\right) \\ & + \sum_{i=1}^{n-1} (-1)^i f\left(a_1 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_n\right) \\ & + (-1)^n f\left(a_1 \otimes \cdots \otimes a_{n-1}\right) a_n\,. \end{aligned}$$

By additivity of $M \otimes_{A^e} -$ and $\mathrm{Hom}_{A^e}(-, M)$, we see that the above sequences are indeed complexes, thus we can compute their (co)homology.

Definition. — The **Hochschild homology** $\mathrm{HH}_*(A;M)$ of A with coefficients in an A-A-bimodule M is the homology of the Hochschild chain complex, that is

$$\mathrm{HH}_n(A;M) := H_n\left(C_\bullet(A;M)\right) = H_n\left(M \otimes_\Lambda A^{\otimes\bullet}\right)\,,$$

and similarly the **Hochschild cohomology** $\mathrm{HH}^*(A;M)$ is

$$\mathrm{HH}^n(A;M) := H^n\left(C^\bullet(A;M)\right) = H^n\left(\mathrm{Hom}_\Lambda(A^{\otimes\bullet}, M)\right)\,.$$

In case $M = A$, we simply write $\mathrm{HH}_*(A)$, resp. $\mathrm{HH}^*(A)$.

Example. — Let $A = \Lambda = k$ be a field. We compute $\mathrm{HH}_*(k)$ by following the definition: since $k \otimes k \simeq k$ via the multiplication, then we get the boundary maps

$$d_n(x_0 \otimes \cdots \otimes x_{n+1}) = \sum_{i=0}^{n+1} (-1)^i x_0 x_1 \dots x_{n+1} = (x_0 x_1 \dots x_{n+1}) \sum_{i=0}^{n+1} (-1)^i$$

and hence the bar resolution turns out to be

$$\mathrm{Bar}(k): \qquad \dots \longrightarrow k \xrightarrow{\sim} k \xrightarrow{0} k \xrightarrow{\sim} k \xrightarrow{0} k \longrightarrow 0\,.$$

Thus the only non trivial cohomology group is $\mathrm{HH}_0(k) = k$. Dually, if we want to compute the Hochschild cohomology we notice that $\mathrm{Hom}_k(k^{\otimes n}, k) \simeq k$ via the isomorphism $f \mapsto f(1 \otimes \cdots \otimes 1)$, and by composing with multiplications one gets

$$C^\bullet(k;k): \qquad 0 \longrightarrow k \xrightarrow{0} k \xrightarrow{\sim} k \xrightarrow{0} k \xrightarrow{\sim} k \xrightarrow{0} \dots\,,$$

thus it holds $\mathrm{HH}^0(k) = k$ and $\mathrm{HH}^n(k) = 0$, for $n > 0$.

As the previous Example shows, computing the Hochschild cohomology of a Λ-algebra by using the bar resolution may be quite tedious and difficult; the next Proposition shows us a way to overcome this issue. We first state an easy

II.4. Lemma. — *If P_1 and P_2 are projective A-modules, then so is $P_1 \otimes_A P_2$.*

Proof. We have the natural isomorphism of functors

$$\mathrm{Hom}_A(P_1 \otimes_A P_2, -) \simeq \mathrm{Hom}_A(P_1, \mathrm{Hom}_A(P_2, -))\,,$$

from which we deduce that the left hand side is an exact functor. □

II.5. Proposition. — *Assume the algebra A is projective as a Λ-module. For every A-A-bimodule M we have canonical isomorphisms*

$$\mathrm{HH}_*(A;M) \simeq \mathrm{Tor}_*^{A^e}(M, A)\,, \quad \mathrm{HH}^*(A;M) \simeq \mathrm{Ext}^*_{A^e}(A, M)\,.$$

Proof. Since by the Lemma II.4 the n-fold tensor product $A^{\otimes n}$ is a projective Λ-module, then $A^{\otimes n+2}$ is projective over A^e. This means that $\mathrm{Bar}(A) \to A \to 0$ is a *projective resolution* of A as an A^e-module, and hence computing Hochschild homology (resp. Hochschild cohomology), is the same as computing the left derived functor of $M \otimes_{A^e} -$, in other words $\mathrm{Tor}_*^{A^e}(M, -)$ (resp. is the right derived functor of $\mathrm{Hom}_{A^e}(-, M)$, i.e. $\mathrm{Ext}^*_{A^e}(-, M)$). □

Example. — Let $\Lambda = k$ be a field and consider $A = k[t]$. We compute $\mathrm{HH}_*(k[t])$. Since $k[t]$ is commutative, $A^{op} = A$ and hence $A^e \simeq k[x, y]$, where $k[x, y]$ has the bimodule structure over $k[t]$ given by

$$p(t) \cdot f(x, y) := p(x) f(x, y)\,, \quad f(x, y) \cdot q(t) := f(x, y) q(y)\,,$$

and vice versa $k[t]$ is a $k[x, y]$-module via the structure map $f(x, y) \mapsto f(t, t)$. Thus, we may consider the following free resolution of $k[t]$, seen as a module over $k[x, y]$:

$$0 \longrightarrow k[x, y] \xrightarrow{(x-y)\cdot} k[x, y] \longrightarrow k[t] \longrightarrow 0\,.$$

After tensoring the truncated resolution with $k[t]$, the multiplication by $(x-y)$ becomes the zero map, thus from

$$0 \longrightarrow \underbrace{k[x,y] \otimes_{k[x,y]} k[t]}_{\simeq k[t]} \xrightarrow{0} \underbrace{k[x,y] \otimes_{k[x,y]} k[t]}_{\simeq k[t]} \longrightarrow 0\,,$$

we conclude that

$$\mathrm{HH}_n(k[t]) = \begin{cases} k[t]\,, & \text{if } n = 0,1\,; \\ 0\,, & \text{if } n \geq 2\,. \end{cases}$$

Example. — Let $\Lambda = k$ be a field and $m \geq 2$. We now show how to compute the Hochschild cohomology of a *truncated polynomial algebra* $A = k[t]/(t^m)$. Define elements

$$u := t \otimes 1 - 1 \otimes t\,, \quad w := \sum_{i=0}^{m-1} t^{m-1-i} \otimes t^i\,,$$

then consider the 2-cyclic free resolution

$$\ldots \xrightarrow{u\cdot} A^e \xrightarrow{w\cdot} A^e \xrightarrow{u\cdot} A^e \xrightarrow{w\cdot} A^e \xrightarrow{u\cdot} A^e \longrightarrow A \longrightarrow 0\,.$$

One can show the above sequence is exact either by a straight computation, or by showing that the following left A-linear maps define a contracting homotopy: set $s_{-1}(1) = 1 \otimes 1$ and for $k \geq 0$ take

$$s_{2k}(1 \otimes x^j) = -\sum_{h=1}^{j} x^{j-h} \otimes x^{h-1}\,, \quad s_{2k+1}(1 \otimes x^j) = \delta_{j,m-1} \otimes 1\,,$$

where $\delta_{i,j}$ is the Kroeneker delta.

After applying $\mathrm{Hom}_{A^e}(-, A)$ and identifying $\mathrm{Hom}_{A^e}(A^{\otimes n}, A) \simeq A$, one gets the complex

$$\ldots \xleftarrow{0} A \xleftarrow{mx^{m-1}\cdot} A \xleftarrow{0} A \xleftarrow{mx^{m-1}\cdot} A \xleftarrow{0} A \longleftarrow 0\,,$$

thus one can compute the Hochschild cohomology, being careful to distinguish between the two cases:

- if $\mathrm{char}(k)$ divides m, then every coboundary map in the above complex is trivial and hence $\mathrm{HH}^n(A) \simeq A$ for every $n \geq 0$;
- if $\mathrm{char}(k)$ does *not* divide m, then we see that

$$\mathrm{HH}^n(A) = \begin{cases} A\,, & \text{if } n = 0\,; \\ (t)/(t^m)\,, & \text{if } n \text{ is odd}\,; \\ k[t]/(t^{m-1})\,, & \text{otherwise}\,. \end{cases}$$

Remark (Interpretation of the Hochschild cohomology in low degree). — Given a bimodule M over A, in virtue of the isomorphism (II.3.1) we may identify Hochschild 0-cochains with elements of M:

$$M \simeq \mathrm{Hom}_{A^e}(A \otimes_\Lambda A, M)\,, \quad m \longmapsto (f : 1 \otimes 1 \mapsto m)\,,$$

thus $m \in M$ is a 0-cocycle if $ma - am = 0$, for every $a \in A$; we deduce that the zeroth Hochschild cohomology is the **submodule of invariants** of M:

$$\mathrm{HH}^0(A; M) = \{\, m \in M \mid \forall_{a \in A}\, ma = am \,\}\ .$$

In particular, if $M = A$ we recover the **centre** of the algebra A, i.e. $\mathrm{HH}^0(A) = Z(A)$.

By using $\mathrm{Hom}_{A^e}(A^{\otimes 3}, M) \simeq \mathrm{Hom}_k(A, M)$, a 1-cocycle is a k-linear map $f : A \to M$ such that

$$af(b) - f(ab) + f(a)b\,, \quad \text{for every } a, b \in A\,;$$

on other words, f satisfies the **Liebniz rule**

$$f(ab) = af(b) + f(a)b\,.$$

Thus, the Hochschild 1-cocycles $Z^1(C^\bullet(A; M))$ is the module of Λ-derivations from A into M, written $\mathrm{Der}_\Lambda(A, M)$. A derivation of the form $\partial_0 m$, for some $m \in M$, is called an **inner derivation** and by quotienting these derivations out, we obtain the module of **outer derivations**:

$$\mathrm{HH}^1(A; M) = \mathrm{OutDer}_\Lambda(A, M)\,.$$

In case $\Lambda = k$ is a field, outer derivations form a vector space which may be endowed with a Lie braket $[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$; then, we see that $\mathrm{HH}^1(A; M)$ has a natural **Lie algebra** structure.

As often happens with cohomology in degree 2, the module $\mathrm{HH}^2(A; M)$ classifies some particular A-A-bimodule extensions, called **Hochschild extensions**. One can show that $\mathrm{HH}^3(A; M)$ classifies other mathematical objects, namely c**rossed bimodules**. These spaces play an important role in the *deformation theory of algebras*.

Chapter III.

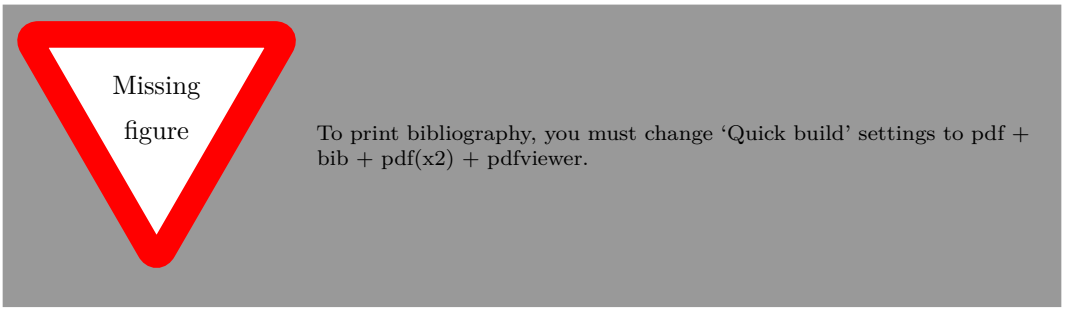
Spherical objects

III

Monoid action on a category

The aim of this paragraph is to describe how to de

fine by generators and relations an action of the braid group on a category; then we will apply this result to the derived category of a scheme $\mathbf{D}^b(X)$.



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